SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1514

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct

I P.B.VAMSI KRISHNA [192525147] (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: PB VAMSI

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

C01-AT2 - P.B.VAMSI KRISHNA 19252514

1) CLOUD COMPUTING FUNDAMENTALS

CLOUD COMPUTING

Definition

cloud computing is the delivery of computing Resource (servers, storage and etc). over the internet instead of using local computer or physical servers

characterstics

* on-demand self service
* Broad Network access
* Resource Pooling
* Rapid elesticity

Benifits

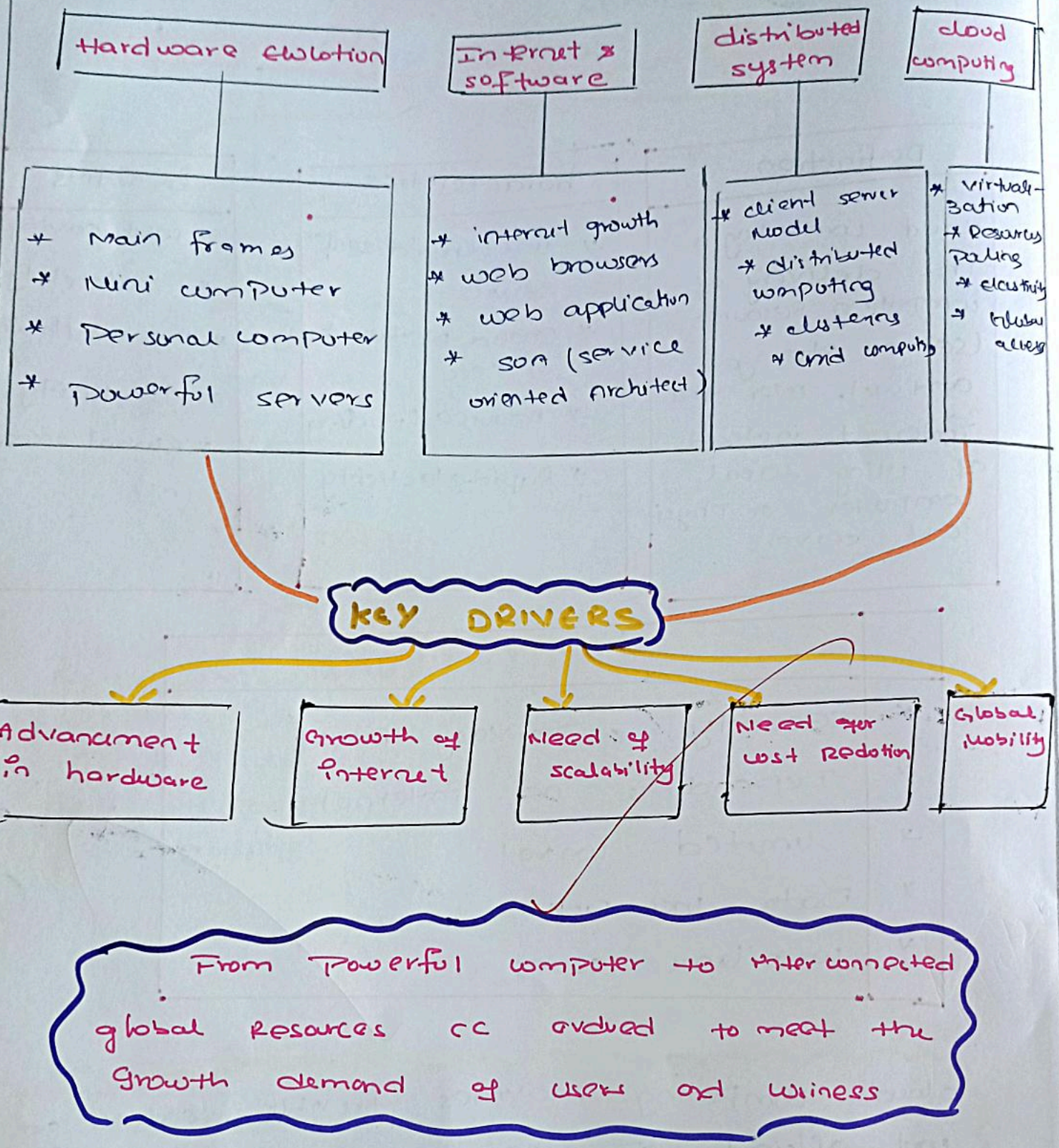
* cost saving
* scalability
* high availability
* Flexibility
* Global access

LIMITATIONS

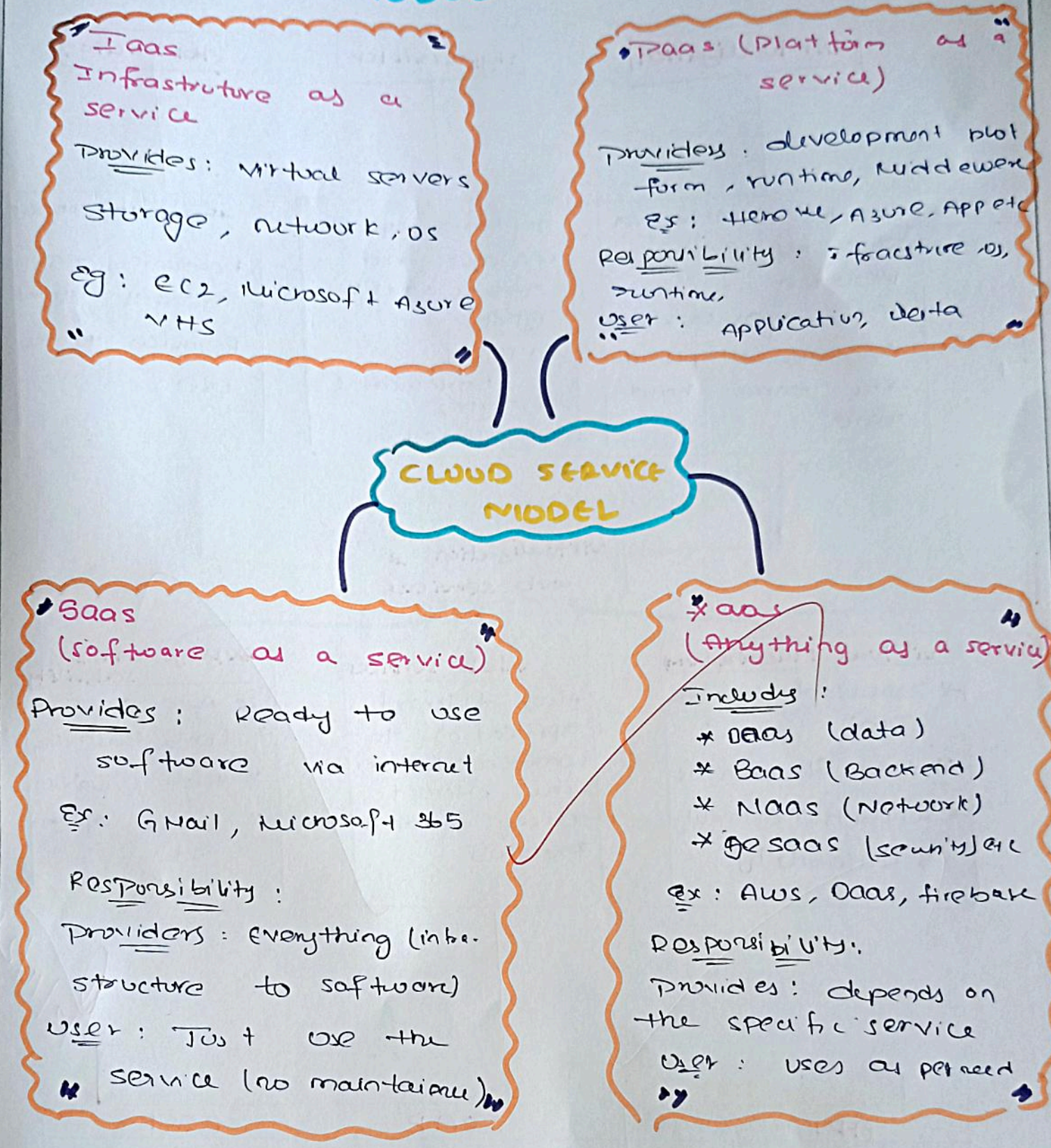
* security & Privacy Risks
* Dependence on internet
* limited control
* Data loss Risk
* compilance issues

cloud computing provides flexible, scalable and cost efficient IT Resources over the internet but it also comes with security privacy and control challenges

2. Evolution of cloud computing



3. SERVICE MODEL



4) VIRTUALISATION AND WEB SERVICES

Virtualization

Creates a virtual version of a physical resource
Ex: servers, OS, storage
to run multiple on the same hardware

Hypervisior

Software that manages and controls VMs:
Types
(i) Type 1
Bare metal
(ii) Type 2
hosted

Abstraction

Hide the physical hardware and presents virtual resource to the user.

virtualization & web services

virtual Machine (VM)

* Isolated environment
* Has its own OS
* Supports Portability & Scalability

Web services

Allos different application to communicate over the internet using standard protocol

Deployement & scalability

* Auto scaling
* Load balancing
* High Availability
* cloud deployment.

13/10/26

API

set of rules access of services

SOAP

protocol for structured XML messaging

REST

simple, light weight web services.

virtualization creates fexible and Efficient use of Resource while web services ed integration between Application in the cloud.

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks: